

Clone Catch Game Repository

October 9, 2026 · <https://chatgpt.com/share/6ac877d7-3cd4-83ee-ae51-036e1be81a2d>

YOU

Catch the Falling Objects Lab This project is a single-topic Catch game using **Pygame**. It introduces students to collision timing, a classic list-mutation pitfall, boundary handling, spawn pacing, and temporary power-ups, using a small, readable object-oriented codebase. --- ## What's Provided A working Catch game with: - A basket the player moves left and right along the bottom of the screen - Objects that spawn at the top at random x positions and fall straight down at a fixed interval - A running score, a miss counter, and a game-over state once too many objects are missed It has **one deliberate bug** (with two distinct symptoms) and **three features** left for you to build. You are expected to **analyze**, **interact with an AI assistant**, and **complete/fix** the game to make it fully functional and more interesting. ### **Use an LLM** (e.g. ChatGPT or Claude) as your debugging and pair-programming partner for this lab. --- ## Getting Started ### Setup 1. Make sure you have Python 3.10+ installed. 2. Install dependencies:

```
bash
pip install -r requirements.txt
```

3. Run the game:

```
bash
python main.py
```

Controls: Left/Right arrows to move the basket. --- ## Tasks to Complete Each task must be completed using an iterative process involving LLM suggestions and your critical code review. ### Task 1: Fix the falling-object collision bug > The basket is supposed to catch an object only once it has actually > fallen down to basket height. In the current build, `is_caught` (in > `game/collision.py`) only checks whether the object's x position is > within the basket's left/right edges - it never checks height at > all. That means an object can score the instant it spawns at the top > of the screen, as soon as the basket happens to be under its x > position, however far away it still is. Separately, the loop in > `update()` (in `game/game_engine.py`) that checks for catches removes > objects from the same list it's iterating over - a classic Python > mistake that can silently skip checking an object on a given frame > when two catchable objects are next to each other in the list. Fix > both: catches should require the object to actually be at basket > height, and the catch-checking loop should not skip entries. ### Task 2: Improve basket control > The basket can currently be moved half off either edge of the > screen, since its boundary check doesn't account for its own width. > Fix the boundary handling so the

basket always stays fully on > screen, and make sure movement feels smooth and responsive even when > a direction key is held down continuously. #### Task 3: Implement controlled object spawning > Improve the spawning system so objects appear at varied horizontal > positions and at varied intervals, rather than a fixed interval with > a plain random x each time. Make sure objects never spawn outside > the playable area, avoid repeatedly spawning in the same spot, and > cap how many objects can be on screen at once so the game never > becomes unplayable. #### Task 4: Add a temporary basket speed boost > Add a speed-boost mechanic the player can activate during play that > temporarily increases the basket's movement speed. Show a clear > indication while the boost is active, and have the basket > automatically return to normal speed once it expires. --- ## Expected Behavior - An object should only be caught once it has genuinely reached the basket - not the moment its x position happens to line up with the basket, however high up it still is. - If two catchable objects are near each other, both should be credited - neither should be silently skipped. - Holding Left or Right continuously should never let the basket move off either edge of the screen. - Objects should spawn at varied positions and timing, never overwhelming the screen or repeating the same spot constantly. - The speed boost should be clearly visible while active and wear off on its own. --- ## Folder Structure

```
catch-falling/
|—— main.py
|—— requirements.txt
|—— game/
|   |—— game_engine.py
|   |—— basket.py
|   |—— falling_object.py
|   |—— collision.py
|   |—— renderer.py
|—— README.md
```

--- ## Submission Checklist Submission is only the following three things: - [] A 10-second video of gameplay **before** your changes, showing the bug/broken behavior - [] A 10-second video of gameplay **after** your changes, showing the bug fixed and the new features working - [] The Chat/LLM used page link, with the complete chat history-----
[SETAPESU26/07_catch_falling](#) can you access this git link help me solve this assignment step by step start from cloning the repo

YOU

```
C:\Users\grees\07_catch_falling>git clone GREESHMA-K77/07\_catch\_falling.git Cloning into
'07_catch_falling'... remote: Enumerating objects: 14, done. remote: Counting objects: 100%
(5/5), done. remote: Compressing objects: 100% (5/5), done. remote: Total 14 (delta 0), reused
0 (delta 0), pack-reused 9 (from 1) Receiving objects: 100% (14/14), 5.32 KiB | 605.00 KiB/s,
done. C:\Users\grees\07_catch_falling>cd 07_catch_falling
C:\Users\grees\07_catch_falling\07_catch_falling>
```

YOU

```
C:\Users\grees\07_catch_falling\07_catch_falling>dir Volume in drive C is Windows-SSD
Volume Serial Number is 3C3A-77C0 Directory of
C:\Users\grees\07_catch_falling\07_catch_falling 09-10-2026 09:26 <DIR> . 09-10-2026 09:26
<DIR> .. 09-10-2026 09:26 34 .gitignore 09-10-2026 09:26 <DIR> catch-falling 09-10-2026
09:26 4,751 README.md 2 File(s) 4,785 bytes 3 Dir(s) 306,539,642,880 bytes free
C:\Users\grees\07_catch_falling\07_catch_falling>git status On branch main Your branch is up
to date with 'origin/main'. nothing to commit, working tree clean
C:\Users\grees\07_catch_falling\07_catch_falling>python --version Python 3.11.9
C:\Users\grees\07_catch_falling\07_catch_falling>python -m pip install -r requirements.txt
ERROR: Could not open requirements file: [Errno 2] No such file or directory: 'requirements.txt'
C:\Users\grees\07_catch_falling\07_catch_falling>
```

YOU

```
Directory of C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling 09-10-2026 09:26
<DIR> . 09-10-2026 09:26 <DIR> .. 09-10-2026 09:26 <DIR> game 09-10-2026 09:26 1,005
main.py 09-10-2026 09:26 15 requirements.txt 2 File(s) 1,020 bytes 3 Dir(s) 306,541,412,352
bytes free C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling>python -m pip install -
r requirements.txt Collecting pygame>=2.5.0 (from -r requirements.txt (line 1)) Downloading
pygame-2.6.1-cp311-cp311-win_amd64.whl.metadata (13 kB) Downloading pygame-2.6.1-
cp311-cp311-win_amd64.whl (10.6 MB)
```

```
10.6/10.6 MB 891.2 kB/s 0:00:12 Installing collected packages: pygame Successfully installed
pygame-2.6.1 C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling>
```

YOU

```
C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling>type game\collision.py """
collision: figures out whether a falling object is within the basket. """ def is_caught(basket_rect,
obj): return basket_rect.left <= obj.x <= basket_rect.right
C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling>type game\game_engine.py """
GameEngine: owns the basket and all falling objects. Starter version: basket movement and
spawning both work at a basic level (Tasks 2 and 3 ask you to improve them), there's no speed
boost yet (Task 4 builds it from scratch), and catch detection has two known bugs (see
game/collision.py and the catch-checking loop below) that Task 1 asks you to fix. """ import
random import pygame from game.basket import Basket from game.falling_object import
FallingObject from game.collision import is_caught from game.renderer import WIDTH,
HEIGHT SPAWN_INTERVAL_FRAMES = 50 MAX_MISSES = 5 class GameEngine: def
__init__(self): self.basket = Basket(x=WIDTH / 2, y=HEIGHT - 30) self.objects = []
self.frames_until_spawn = 0 self.score = 0 self.misses = 0 self.game_over = False def
_spawn_object(self): x = random.randint(20, WIDTH - 20)
self.objects.append(FallingObject(x=x, y=-14, speed=3)) def handle_input(self, keys_pressed):
if self.game_over: return if keys_pressed[pygame.K_LEFT]: self.basket.x -= self.basket.speed
if keys_pressed[pygame.K_RIGHT]: self.basket.x += self.basket.speed # Boundary handling:
only clamps against the screen edges, not # accounting for the basket's own width - it can
hang half off # either side of the screen. self.basket.x = max(0, min(WIDTH, self.basket.x)) def
handle_keydown(self, key): if self.game_over and key == pygame.K_r: self.__init__() def
update(self): if self.game_over: return self.frames_until_spawn -= 1 if self.frames_until_spawn
<= 0: self._spawn_object() self.frames_until_spawn = SPAWN_INTERVAL_FRAMES for obj in
self.objects: obj.update() basket_rect = self.basket.get_rect() for obj in self.objects: # BUG:
mutating this list while iterating over it if is_caught(basket_rect, obj): self.score += 1
self.objects.remove(obj) missed = [o for o in self.objects if o.is_past_bottom(HEIGHT)] if
missed: self.objects = [o for o in self.objects if not o.is_past_bottom(HEIGHT)] self.misses +=
len(missed) if self.misses >= MAX_MISSES: self.game_over = True def draw(self, surface,
font): from game import renderer renderer.draw_scene(surface, self.basket, self.objects)
renderer.draw_text(surface, font, f"Score: {self.score}", (10, 10)) renderer.draw_text(surface,
font, f"Misses: {self.misses}/{MAX_MISSES}", (10, 36)) if self.game_over:
renderer.draw_banner(surface, font, f"Game Over! Final score: {self.score}. Press R to
restart.") C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling>-----while playing
game i was not able to move sidebar
```

YOU

no basket moved i used arrows now give next steps

YOU

pygame 2.6.1 (SDL 2.28.4, Python 3.11.9) Hello from the pygame community.

<https://www.pygame.org/contribute.html>

```
C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling>type main.py """ Catch the
Falling Objects (Lab Starter) Run with: python3 main.py Controls: Left/Right arrows to move
the basket. """ import pygame from game.game_engine import GameEngine from
game.renderer import WINDOW_SIZE def main(): pygame.init() screen =
pygame.display.set_mode(WINDOW_SIZE) pygame.display.set_caption("Catch the Falling
Objects") clock = pygame.time.Clock() font = pygame.font.SysFont("consolas", 22) engine =
GameEngine() running = True while running: for event in pygame.event.get(): if event.type ==
pygame.QUIT: running = False elif event.type == pygame.KEYDOWN:
engine.handle_keydown(event.key) keys = pygame.key.get_pressed()
engine.handle_input(keys) engine.update() engine.draw(screen, font) pygame.display.flip()
clock.tick(60) pygame.quit() if __name__ == "__main__": main()
C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling>type game\basket.py """ Basket:
the player-controlled catcher at the bottom of the screen. """ import pygame class Basket: def
__init__(self, x, y, width=90, height=24, speed=5): self.x = x self.y = y self.width = width
self.height = height self.speed = speed self.boosted_frames = 0 self.normal_speed = speed
self.boost_speed = speed def get_rect(self): return pygame.Rect( int(self.x - self.width / 2),
int(self.y - self.height / 2), self.width, self.height, )
C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling>type game\falling_object.py """
FallingObject: a simple object that falls straight down. """ class FallingObject: def __init__(self,
x, y, radius=14, speed=3, color=(230, 140, 60)): self.x = x self.y = y self.radius = radius
self.speed = speed self.color = color def update(self): self.y += self.speed def
is_past_bottom(self, height): return self.y - self.radius > height
C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling>
```

YOU

no game ran perfectly --- C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling>python
[main.py](https://www.pygame.org/contribute.html) pygame 2.6.1 (SDL 2.28.4, Python 3.11.9) Hello from the pygame community.

<https://www.pygame.org/contribute.html>

C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling>

YOU

```
C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling>git status
On branch main
Your branch is up to date with 'origin/main'.
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   game/basket.py
    modified:   game/game_engine.py
no changes added to commit (use "git add" and/or "git commit -a")
```

```
C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling>git add game/basket.py game/game_engine.py
C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling>git commit -m "Add temporary basket speed boost"
[main d99088f] Add temporary basket speed boost
2 files changed, 40 insertions(+), 7 deletions(-)
```

```
C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling>git push origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 966 bytes | 483.00 KiB/s, done.
Total 6 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
To GREESHMA-K77/07\_catch\_falling.git
5be758f..d99088f main -> main
C:\Users\grees\07_catch_falling\07_catch_falling\catch-falling>
```

YOU

only 4 tasks

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